LAB ASSIGNMENT 7
Week 7
(Generics & Utility Classes)

Q1. Write a Java program to create a generic method that calculates the sum of an array of numeric values. Use a bounded type parameter such as T extends Number and the doubleValue( ) method to calculate the sum. Test the method using arrays of integers, floats, and doubles.

Q2. Write a Java program that stores a student's email address in an Optional<String> object. Perform the following operations:

1. Create an Optional containing an email address.
2. Create an empty Optional.
3. Display the email using ifPresent( ).
4. Use orElse( ) to provide a default email address.
5. Use orElseThrow( ) to handle a missing email value.

Q3. Create a generic class Result<T extends Number> to store marks of a student.
The program should:

1. Store marks using a generic type.
2. Calculate the percentage or average using numeric conversion methods.
3. Determine whether the student has passed based on a specified threshold.
4. Display the result using Optional if the marks are available.
5. Use Objects.requireNonNull( ) to validate required inputs.

Q4. Write a Java program to manage an event using LocalDate and LocalTime.
The program should:

1. Accept an event name, event date, and start time.
2. Display the event details.
3. Check whether the event date is in the future, today, or in the past.
4. Calculate the number of days remaining until the event.
5. Calculate the event's end time if the event lasts for two hours.
6. Use Optional<String> to handle an optional venue name and display "Venue not specified" when it is absent.

Q5. Develop a generic utility class containing the following methods:

1. isEqual(T a, T b) — checks whether two objects are equal.
2. getDefault(Optional<T> value, T defaultValue) — returns the contained value or a default.
3. max(T a, T b) — returns the larger of two comparable objects.
4. isNull(Object obj) — checks whether an object is null using the Objects class.

Demonstrate the methods using strings, integers, and other suitable objects.